When Does Forecast Accuracy Predict Inventory Cost?

Evidence from a Maritime Spare-Parts Fleet

Nishchay Patel

September 8, 2026

Abstract

Whether forecast accuracy predicts inventory cost is usually answered by correlating the two across methods. On 15,348 maritime spare-parts series under an (s, S) policy, with the operator's deployed policy as baseline, that statistic fails twice. It is confounded: MAE is minimised at the median of demand, zero for 77.3 % of these series, so ranking methods by MAE ranks them by forecast level, six copies of one forecaster differing only by a scale factor reproduce the full regime pattern, correlating at +1.000 where holding cost dominates. The regime boundary follows from the newsvendor fractile at a derivable unit price, here \$1,600. The statistic is also fragile, changing sign across nine defensible evaluation designs while per-SKU statistics barely move. Scoring the critical fractile inherits the confound; scoring the fractile the replenishment decision reads escapes it. On M5 the zero-median condition holds for 10.1 % of series, helping reconcile the conflicting evidence.

Keywords: intermittent demand; spare-parts inventory; cost-aware forecast evaluation; cost regime; foundation models; maritime operations.

1 Introduction

Forecasting models are judged by accuracy metrics such as MAE, MASE and RMSSE. In spare-parts management the model is not the end of the line: it feeds an (s, S) inventory policy, and what that policy costs is holding cost plus whatever stockouts cost. Whether the two criteria agree has been asked for two decades (Syntetos & Boylan 2006; Teunter et al. 2010; Prak & Teunter 2019; Kolassa 2016), and the answers have not been consistent.

Recent work sharpens the question from two directions. Petropoulos, Wang & Disney (2019) compared inventory performance across the M3 methods, and Theodorou, Spiliotis & Assimakopoulos (2025) found on the M5 data that the accuracy-to-cost link is conditional on the cost structure: accuracy is more relevant when holding cost is comparable to or larger than lost-sales cost, and where lost-sales cost dominates the preferable method may not be the most accurate one. Separately, it is known that point-error metrics misbehave on mostly-zero series: MAE rewards a flat zero forecast, which motivated MASE (Hyndman & Koehler 2006), and Wallström & Segerstedt (2010) and Kolassa (2016, 2020) argue the problem runs deeper than any one metric.

This paper connects the two observations and pushes the conclusion further than either. We work on the spare-parts operation of a multi-brand cruise group, three brands and 40 vessels, around 220k SKUs with quarterly history, with the (s, S) parameters the operator actually runs as a baseline. On this data, under policies that read the forecast through its mean, the regime pattern is not a fact about forecast quality at all. It is what the metric-level confound looks like after it passes through a cost function; its boundary is derivable from the cost parameters alone; it reproduces in full on six copies of one forecaster that differ only by a scale factor; and the fleet-level statistic that exhibits it flips sign under evaluation-design choices that studies in this literature rarely report.

Contributions.

- **MAE is confounded with forecast level, by identity rather than by tendency.** For a flat forecast, MAE is minimised at the median of realised demand. That median is zero for 77.3 % of these SKUs, and on such a SKU MAE is strictly increasing in the forecast, so ranking forecasters by MAE ranks them by level. Per SKU the level-MAE rank correlation has median +1.000, with 78.6 % of SKUs at exactly +1, and six scaled copies of one forecaster, which contain identical information about demand, reproduce the full regime pattern at +1.000 where holding cost dominates.
- **The regime boundary has a single-period derivation.** The cost-minimising forecast is the quantile at the newsvendor critical fractile $q^* = C_u/(C_u + C_o)$, which crosses the 0.5 fractile MAE targets at a derivable unit price, \$1,600 here, and the sign of the accuracy-cost correlation differs across that price as predicted. Where the zero-median condition binds, the regime pattern is therefore the confound's fingerprint: real, predictable from the cost parameters, and not evidence that better forecasts cost less. On M5 the condition binds for one series in ten, so the boundary Theodorou et al. (2025) report cannot be wholly this identity at work.
- **The fleet-level statistic is fragile as an instrument.** The correlation between method-mean accuracy and method-mean cost rests on eleven points. Across nine defensible evaluation designs it runs from −0.103 to +1.000 in the holding-dominated group, and dropping one model moves it from +0.591 to +0.455. Per-arm bootstrap intervals are wide enough that the extreme designs overlap, so the instrument is underpowered as well as unstable, and either reading disqualifies it. The per-SKU τ, computed over 15,348 SKUs, stays between +0.333 and +0.435 in every design.
- **The obvious remedy fails, and the working one is narrower than expected.** Scoring at each SKU's own q^* is confounded with level just as badly in the opposite direction (per-SKU pinball-level rank correlation median −1.000). Under a policy that consumes the whole predictive distribution the confound is escapable, but only at the fractile the decision reads. On the common SKU set the paired pinball-minus-MAE difference is +0.024 with an interval touching zero, directional rather than established, while a generic five-quantile CRPS is strictly worse than MAE (paired difference −0.107, interval excluding zero). The comparison rests on the four models that emit quantiles.
- **The severity is measurable in advance.** On the public M5 data the level-MAE statistic has median +0.194 because the analytic condition holds for only 10.1 % of M5 series against 77.3 % here; conditioning on it, the mechanism reappears (+0.614 against +0.157). The confound is a property of how zero-dense the evaluation window is, not of intermittent demand as such, and competition-data and spare-parts studies sit on opposite sides of it.
- **Secondary results.** The within-SKU and fleet-level statistics answer different questions and disagree (+0.387 against +0.018). A zero-shot Chronos-T5 model is the most accurate and the cheapest point estimate of eleven forecasters, though separable from almost none of them. Ten of eleven model-driven policies beat the operator's deployed parameterisation,

with a bootstrap interval on the gap of $[-60.2\,\%, -41.0\,\%]$. And deliberately scoring against right-censored quarters does not reorder the forecasters.

Section 2 places the work; Section 3 covers data; Section 4 methods; Section 5 results; Sections 6 to 8 discuss, bound and conclude. A reproducibility note records the analysis history, including several corrections.

2 Related work

Intermittent-demand forecasting. Croston (1972) smoothed demand size and inter-demand interval separately. Syntetos & Boylan (2005) corrected its inversion bias, giving SBA; Teunter, Syntetos & Babai (2011) smoothed the demand probability instead (TSB); Nikolopoulos et al. (2011) introduced temporal aggregation (ADIDA/IMAPA); Kourentzes (2014) added parameter tuning. MAPE is undefined on mostly-zero series and plain MAE rewards a flat zero forecast, so Hyndman & Koehler (2006) standardised the scaled error (MASE).

Accuracy versus inventory performance. Syntetos & Boylan (2006) showed empirically that accuracy and stock-control rankings can diverge; Teunter et al. (2010) and Prak & Teunter (2019) developed the gap analytically; Teunter & Duncan (2009) showed the accuracy ranking of intermittent-demand methods depends on the error measure chosen, and Syntetos, Nikolopoulos & Boylan (2010) proposed judging forecasts by accuracy-implication metrics rather than accuracy itself. On the metric side, Wallström & Segerstedt (2010) found that error measures for intermittent series disagree with each other and with stock-control outcomes, Kolassa (2016) argued point-error metrics are the wrong target for count data, and Kolassa (2020) showed the "best" point forecast is a property of the error measure chosen. Petropoulos, Wang & Disney (2019) measured inventory performance across M3 methods, and Theodorou, Spiliotis & Assimakopoulos (2025) found on M5 that the accuracy-cost relationship depends on the balance between holding and lost-sales cost. This paper argues those two lines are reporting the same phenomenon: the regime dependence is what the metric-level confound produces once it passes through a cost function, its boundary is derivable from the cost parameters, and the fleet-level statistic that exhibits it is fragile to evaluation design.

Probabilistic forecasting and calibration. Gneiting et al. (2007) formalised calibration and sharpness. We use predictive quantiles to set safety stock and report the coverage they achieve.

Global and foundation-model forecasting. Cross-learning across series (Salinas et al. 2020; Bandara et al. 2020; Makridakis et al. 2022) set up the zero-shot foundation models, of which Chronos (Ansari et al. 2024) is one. Vandeput (2020) cautions that machine learners over-predict when most observations are zero. Head-to-head evaluations of a foundation model against Croston-family methods on intermittent spares, scored by inventory cost rather than accuracy alone, appear to be scarce.

Inventory policy. The (s, S) machinery is textbook (Silver, Pyke & Thomas 2017; Snyder & Shen 2019).

3 Data and deployed pipeline

3.1 Source, scope and sampling

The operator runs four maintenance-management instances, three legacy and one migrated. We extract spare-part consumption, the parts master, the forecast-analysis table and the deployed (s,

S) parameter table, restricted to the cruise group's operating segment. Anonymisation is keyed and salted: every identifying field (brand, vessel, SKU, part type, vendor, department) becomes a namespaced surrogate, consistent across tables. Free-text fields are dropped. Dates, quantities, prices, lead times, the demand-pattern segmentation, ABC class and criticality are preserved.

The analysis runs on a **stratified sample of 15,348 SKUs** drawn from the 219,783-SKU forecast panel by equal allocation across demand-pattern and ABC cells, with a per-cell cap of 1,579. Equal allocation buys precision in the small, high-value cells that a proportional sample would barely reach, and it is the right design for a paired model comparison. It is the wrong design for reading a raw mean as a fleet quantity: sampling fractions run from 1.0 % to 100 %, so the design weight $w_h = N_h/n_h$ spans 1.00 to 96.12, and the single Intermittent × C cell is 69 % of the panel against 10 % of the sample.

We therefore report both. Model comparisons, which are paired within SKU, use the sample unweighted. Every quantity stated as a property of the fleet is also reported as a Horvitz–Thompson estimate carrying w_h, with a bootstrap that resamples within strata (Section 5.10). Kish's effective sample size under those weights is 3,127, not 15,348, and we quote it wherever a weighted interval is given.

3.2 Right-censoring and the evaluation window

Each SKU carries 28 quarterly observations. The fraction with nonzero demand holds at about 5 to 7 % through panel quarter 20, then falls to roughly 1 % by quarters 25 to 28 (Figure 1). This is a reporting lag rather than a demand collapse: the consumption log is stable through the most recent complete calendar quarter and only then drops, because recent transactions have not all been entered. The last panel quarters are incomplete.

We therefore evaluate on **train = panel quarters 1 to 16, test = quarters 17 to 20**, four quarters at roughly 5 % nonzero, matching the training level. Of the 15,348 SKUs, 5,929 (38.6 %) see positive demand in that window. Section 5.9 reports what happens when the test window is deliberately placed in the incomplete region instead.

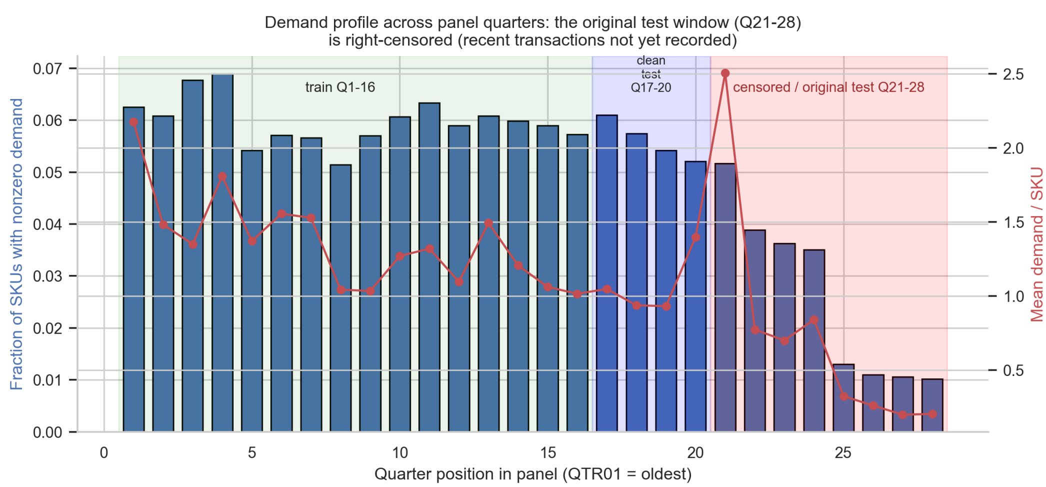

Figure 1: Demand profile across panel quarters. The most recent quarters are incomplete; the evaluation window is Q17–20.

4 Methods

4.1 Forecasters

Eleven models in three tiers. **Classical (seven):** the operator's deployed SBA, SES and two-quarter moving average, plus Croston without bias correction, TSB, ADIDA and IMAPA. **Distributional (two):** an intercept-only Zero-Inflated Poisson (ZIP) and a Hurdle Negative-Binomial (HNB), fit by method-of-moments and turned into predictive quantiles by sampling 1,000 draws. **Global and foundation (two):** a global LightGBM (Ke et al. 2017) with separate occurrence, size and quantile heads, and Chronos-T5-small run zero-shot on the 16-quarter context with 1,000 seeded sample paths.

Two implementation points matter for reproducibility. LightGBM's quantile heads are fit on positive-target rows, so they estimate the size quantile conditional on occurrence; the predictive quantile of demand is the hurdle mixture of that with the occurrence probability, not the conditional quantile gated by an indicator. And LightGBM is non-deterministic across runs when multithreaded unless its seeds, `deterministic`, `force_row_wise` and thread count are all pinned. Both points are elaborated in the reproducibility note, because getting either wrong changes the conclusions.

4.2 (s, S) cost simulator

For each SKU and forecaster we build an (s, S) policy and replay the test-window demand: quarterly review, integer-quarter lead times ($\lceil$days/91.3125$\rceil$, minimum one), and holding cost charged on end-of-quarter positive on-hand. Default configuration: holding 25 %/yr of unit value, ordering \$50 per order, stockout \$100 per unit short. Section 5.8 sweeps all three.

Four modelling choices shape these results and are stated explicitly, because each one is consequential on a four-quarter window.

- **Unmet demand is lost, not backordered.** A stockout is charged once per unit short and the shortfall does not carry into the next quarter.
- **Inventory starts saturated** at on-hand $= S$. On this window that is not a neutral choice: 81.8 % of SKUs never place an order in the four test quarters, so for most of the sample the simulation measures the cost of holding an initial position rather than the cost of replenishing. Starting from zero instead raises mean cost by about 5 % but drops mean fill from 0.965 to 0.830, so the start condition matters far more for service than for cost. The specification grid of Section 5.4 re-runs the whole comparison from four opening positions and three windows, so this choice is tested rather than assumed.
- **Fill rate is 1.0 by definition when a SKU has no realised demand**, which is 61.4 % of the sample. Reported fleet fill rates are therefore mostly definitional; the demand-active mean fill is closer to 0.91. We use fill only to match service between the two policy formulas, never as a headline result.
- **Cost is dominated by holding.** Under the default configuration the split is 90.5 % holding, 9.0 % stockout, 0.5 % ordering. Since holding cost is close to monotone in forecast level, the cost ranking is substantially a ranking of forecast bias, and Section 5.3 should be read with that in mind.

The service level enters only through $z = \Phi^{-1}(\mathrm{SL})$ in the safety-stock term, so it is applied as a cycle-service (Type-1) quantity even though the criticality tiers were specified as availability targets.

Normal-approximation policy. The reorder point is

$$s \;=\; \bar{D}\cdot \mathrm{LT} \;+\; z\sqrt{\mathrm{LT}\cdot\sigma_D^2 + \bar{D}^2\,\sigma_{\mathrm{LT}}^2}\,, \qquad S = s + \mathrm{EOQ},$$

with EOQ from the standard formula. σ_D is held model-agnostic, one training-window standard deviation shared by every forecaster, so the only thing that varies across forecasters is the point forecast $\bar{D}$. This isolates point-forecast quality, at the cost of not testing forecaster-specific uncertainty (Section 7).

Native-quantile policy. For the four models that emit quantiles we set s from the model's own predictive quantile of lead-time demand at the SKU's criticality-tiered service level, then rescale by a bisection-searched multiplier κ until fleet fill matches that model's normal-approximation policy, and compare cost at equal service.

4.3 Metrics and the cost regime

Point accuracy is MAE, with RMSSE as a genuinely independent check. We do not report MASE as a robustness check on the rank statistic: MASE is a strictly positive per-SKU rescaling of MAE, so the per-SKU rank correlation is identical to the MAE one by construction and corroborates nothing.

We report the accuracy-cost relationship at two levels, because they answer different questions. The **within-SKU** statistic is the mean per-SKU Kendall's τ between MAE-rank and cost-rank across the eleven forecasters, with SKU-level bootstrap CIs. The **fleet-level** statistic is the Spearman correlation between each forecaster's mean MAE and its mean cost, across the eleven forecasters. ZIP and HNB share a point forecast (Section 5.1), so the eleven forecasters contribute ten distinct points to this correlation; we keep both and let the tie stand. The second statistic is what a practitioner implicitly uses when choosing one model for the whole fleet.

Cost regime. Under the default configuration, per-unit-quarter holding cost is $(0.25/4)\cdot\text{price}$ and the stockout penalty is a flat \$100, so the two are equal at a unit price of \$1,600. SKUs below that are stockout-dominated; above it, holding-dominated. This gives a direct test of the Theodorou et al. boundary.

5 Results

All results are on the uncensored window: train quarters 1 to 16, test 17 to 20.

5.1 Accuracy

Mean test MAE, best to worst: Chronos 6.525, MA 7.598, SES 7.889, IMAPA 8.201, ADIDA 8.311, HNB 8.595, ZIP 8.595, TSB 8.633, LGBM 8.775, SBA 8.948, Croston 9.107 (Figure 2, x-axis). Chronos wins zero-shot with no fine-tuning. SBA and Croston come last, which fits: a bias-corrected estimator is not built to minimise raw error when most periods are zero.

ZIP and HNB have algebraically identical point forecasts here (both reduce to the training sample mean), so the study contains ten distinct point forecasters, not eleven. We keep both because their predictive distributions differ, which matters in Section 5.7.

5.2 Cost

Mean cost per SKU, cheapest to dearest: Chronos \$2,097, HNB \$2,554, ZIP \$2,555, TSB \$2,564, SBA \$2,581, Croston \$2,585, IMAPA \$2,607, SES \$2,657, ADIDA \$2,691, MA \$2,879, LGBM \$2,936, at fill rates from 0.956 to 0.980 (Figure 2, y-axis).

Chronos is both the most accurate and the cheapest. The second-most-accurate model, the moving average, is the second most expensive. That pair is the whole problem this paper is about, and Section 5.3 explains why it happens.

The ordering is almost entirely unresolvable, and we quantify that rather than noting it in passing. Simulated cost is extremely right-skewed: 3.2 % of SKUs carry 82.3 % of it. Those are not an independently identified tail; they are the holding-dominated group of Section 5.4, since the same unit-price threshold defines both. A paired bootstrap over SKUs ($B = 20{,}000$) gives Chronos a mean of \$2,097 with a 95 % interval of [\$545, \$5,089]. On the paired difference against the cheapest model, **only one of ten models is distinguishable**: LightGBM, at +\$839 $[+212, +1{,}834]$. Chronos against Hurdle-NB is +\$457 $[-232, +1{,}611]$. So Chronos is the cheapest *point estimate* but is not statistically separable from nine of the other ten, and no claim in this paper rests on the cost ordering of adjacent models.

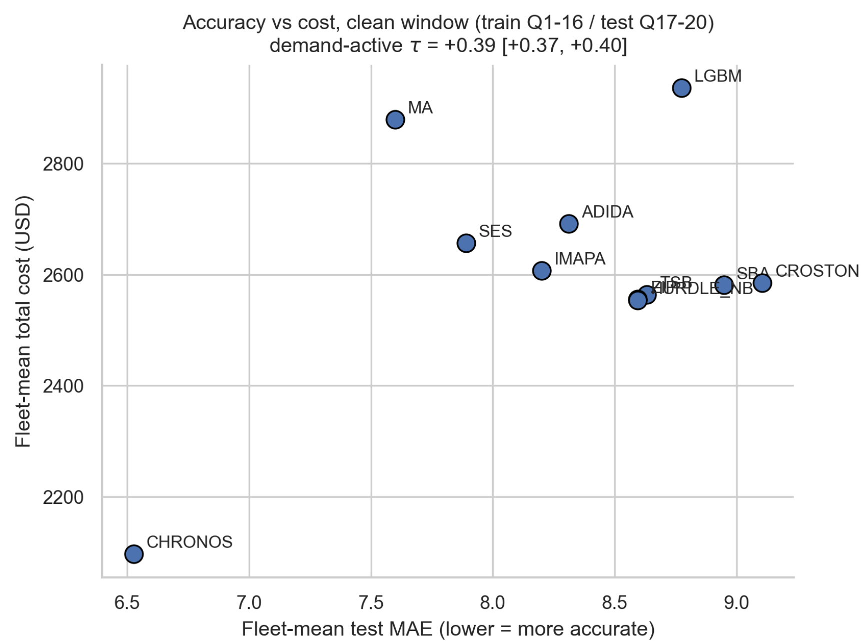

Figure 2: Accuracy (test MAE) versus simulated cost. The most accurate model (Chronos) is also the cheapest; the second (MA) is the dearest.

5.3 MAE is confounded with forecast level

Within-SKU. On a SKU with no realised demand both MAE and holding cost increase with forecast magnitude, so τ is positive by construction; we rely on the demand-active value. Under RMSSE the same statistic falls to $+0.175$ $[+0.154, +0.192]$, so even within-SKU the answer depends on the error metric (Figure 3).

Table 1: Mean per-SKU Kendall's τ between MAE-rank and cost-rank.

SKU subset	n (τ defined)	τ	95 % CI
Demand-active	5,341	+0.387	[+0.369, +0.405]
Zero-demand	9,419	+0.999	mechanical, excluded
Pooled	14,760	+0.778	inflated by zero-demand

Fleet-level. Across the eleven forecasters, Spearman ρ between mean MAE and mean cost is **+0.018** ($p = 0.958$): no relationship at all. The within-SKU and fleet-level statistics are both correct and they disagree, because they ask different questions. Averaging ranks within SKUs measures whether a better forecast for a given part tends to cost less for that part. Correlating model means measures whether a more accurate model is a cheaper model for the fleet, which is the question a practitioner selecting one forecaster is actually asking.

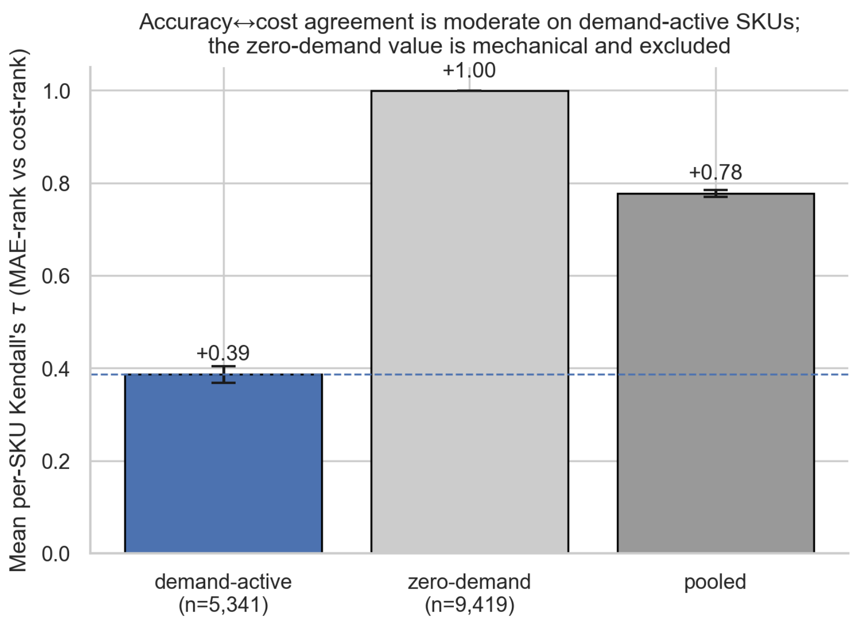

Figure 3: Mean per-SKU Kendall's τ (MAE-rank versus cost-rank), split by demand activity. The zero-demand value is mechanical and excluded.

The mechanism. The two statistics above disagree because neither is measuring forecast quality alone. Start analytically. For a flat forecast $f > 0$ on a window $d_1, \ldots, d_H$,

$$\frac{d\,\mathrm{MAE}}{df} = \frac{\#\{t : f > d_t\} - \#\{t : f < d_t\}}{H},$$

so if more than half the d_t are zero then MAE is strictly increasing in f for every $f > 0$. On such a SKU, ranking forecasters by MAE is *identically* ranking them by forecast level, inverted. This is a property of the metric on zero-dense data, not a tendency that washes out in a larger sample.

The condition binds here. The median of the test window is zero for **77.3 %** of SKUs (61.4 % are entirely zero), and even among demand-active SKUs 41.3 % still have a zero median. Measured per SKU across the eleven forecasters, the rank correlation between forecast level and MAE has

median +1.000, with 78.6 % of SKUs at exactly +1 (n = 14,760). These use 15,348 SKUs rather than eleven model means.

Table 2: Decomposition of the accuracy-cost relationship. In the holding-dominated group MAE and forecast level are perfectly rank-correlated, so nothing survives conditioning on level.

Subset	ρ(MAE, cost)	ρ(MAE, level)	partial	ρ(MAE, hold)	ρ(MAE, stockout)
All	+0.018	+0.982	−0.187	+0.018	−0.509
Stockout-dominated	−0.345	+0.964	+0.548	+0.827	−0.509
Holding-dominated	+0.591	**+1.000**	**+0.000**	+0.591	+0.091

In the holding-dominated group, where the raw correlation is strongest, MAE and forecast level are perfectly rank-correlated and the partial correlation is exactly zero: accuracy and level cannot be separated there at all. The mechanism is visible in the last two columns. Forecast level raises holding cost ($\rho = +0.827$, $p = 0.002$) and lowers stockout cost ($\rho = -0.509$). Which of those two components dominates the total is exactly what the cost regime decides, and that, not forecast quality, is what sets the sign of the accuracy-cost relationship.

The practical consequence is stronger than "accuracy is a weak proxy". Because the confound is structural rather than incidental, MAE cannot serve as a model-selection criterion for these SKUs in *either* regime: where it appears to work it is standing in for level, and where it appears to fail it is also standing in for level, with the sign flipped by the cost structure.

The confound, demonstrated with no information differences at all. Take the deployed SBA forecast and multiply it by $k \in \{0.25, 0.5, 0.75, 1, 1.5, 2\}$. The six variants contain identical information about demand; they differ only in level. On SKUs with a positive base forecast and a zero test median, MAE is strictly increasing in k on **100.0 %** of SKUs, which is the derivative above verified exactly.

Table 3: The scale family $k\times$SBA: identical information, full regime pattern. Stockout-dominated cost is U-shaped in k; holding-dominated cost is monotone.

k	mean MAE	cost, stockout-dom.	cost, holding-dom.
0.25	6.903	$563	$53,117
0.50	7.383	$498	$57,667
1.00	8.916	$470	$66,745
2.00	14.162	$498	$84,991

The fleet-level statistic reproduces the entire regime pattern on this family. ρ(mean MAE, mean cost) across the six variants is **+1.000** where holding cost dominates, and +1.000 pooled, because the holding-dominated group carries 82.3 % of the dollars. In the stockout-dominated group mean cost is U-shaped in k, minimised near $k = 1$, so the correlation there is −0.486 and means nothing: a monotone metric against a non-monotone cost. A statistic that awards a perfect accuracy-to-cost relationship to six copies of the same forecast is not measuring forecast quality. We suggest this scale-family placebo as a standard check for accuracy-versus-cost studies: whatever relationship the pipeline awards to scaled copies is its level effect, and only signal beyond that is evidence about accuracy.

5.4 The regime boundary: derivable, and fragile as an instrument

The empirical split. Splitting at the \$1,600 crossover:

Table 4: The accuracy-cost relationship by cost regime.

Regime	n SKUs	fleet ρ	within-SKU τ	most accurate	cheapest
Stockout-dominated (<\$1,600)	14,861	−0.345	+0.385	Chronos	LGBM
Holding-dominated (≥\$1,600)	487	**+0.591**	+0.491	Chronos	Chronos

A paired bootstrap over SKUs puts intervals on these. The holding-dominated correlation is +0.591 with a 95 % interval of $[+\mathbf{0.49}, +\mathbf{1.00}]$, which excludes zero, so it survives inference that resamples SKUs rather than relying on an eleven-point asymptotic test (whose p was 0.056). The stockout-dominated correlation is −0.345 with an interval of $[-\mathbf{0.76}, +\mathbf{0.44}]$, which includes zero: there is no detectable relationship there, and we do not claim a negative one. Pooled across all SKUs the interval is $[-0.29, +0.75]$, also including zero.

We report these intervals to two decimals deliberately. At $B = 2{,}000$ the holding-dominated lower bound was not stable in the third decimal, ranging from +0.409 to +0.545 across five seeds; at $B = 20{,}000$, which is what we now run, the same sweep spans +0.464 to +0.500. The percentile endpoint of a rank correlation over eleven model means is the fragile statistic here, so alongside it we report the bootstrap share of replicates at or below zero, which is stable: 0.0014 in the holding-dominated group, 0.727 in the stockout-dominated group and 0.447 pooled.

The crossover is derivable. In a newsvendor trade-off with underage cost C_u and overage cost C_o, the cost-minimising order quantity is the demand quantile at the critical fractile $q^* = C_u/(C_u + C_o)$. Here C_u is the \$100 stockout penalty and $C_o = (0.25/4) \times$ price per quarter, so

$$q^*(\text{price}) = \frac{100}{100 + 0.0625\,\text{price}},$$

which equals 0.5 at exactly **price = \$1,600**: the crossover used in Table 4, derived rather than observed. The derivation is single-period and myopic. With a saturated start most units are held for several quarters, which pushes the true break-even price below \$1,600, so we use \$1,600 as the definition of the split rather than as an estimate of the optimum. The data agree: sweeping the split price, the correlation among SKUs above the threshold is positive with a bootstrap zero-share of 0.02 or less from \$100 upward, and disjoint price bands turn significantly positive from the \$194–475 band on (+0.745, zero-share 0.001), so the empirical transition sits near the residence-time boundary (about \$400 if a unit is held for the full four-quarter window), well below the myopic \$1,600.

The mechanism passes a falsification test. If the regime pattern is q^* varying with price against a metric that targets 0.5 everywhere, then a price-proportional penalty, $C_u = (0.25/4) \times \text{price}$, which fixes $q^* = 0.5$ for every SKU, should make the price split stop separating the correlation. It does. Re-costing the same simulations with that penalty gives +0.591 above \$1,600 and +0.582 below, against +0.591 and −0.464 under the published costing. The regime contrast is a property of the penalty structure, not of the parts. MAE implicitly targets the 0.5 fractile for every SKU regardless of price, so it is aligned with the cost structure at exactly one price and misaligned on both sides, which is why the correlation reverses sign instead of merely weakening. The misalignment is severe: median q^* here is **0.981** and 96.8 % of SKUs require $q^* > 0.5$.

Under this design, the split reproduces the boundary Theodorou et al. (2025) report on M5 retail data: accuracy appears to predict cost where holding cost dominates and not where stockouts dominate, and because 96.8 % of these parts are stockout-dominated the pooled correlation is nil. A study that reports one pooled number would conclude accuracy does not matter; a study of high-value slow movers would conclude it does. But Section 5.3 says what this pattern is made of, and the derivation above says where its boundary must sit. The regime split is the level confound passed through a cost function, and reproducing it here, in a different industry, under a different policy class, against a deployed baseline and in a far more intermittent regime, is evidence about the metric and the cost parameters rather than about forecast quality.

How much does the instrument itself move? Everything above is computed over eleven method means under one evaluation design, and eleven points invite the question of what happens under another. We re-ran the full comparison under nine designs, each a combination of an opening inventory position (saturated at S, the published choice; uniform in $[s, S]$; exactly at s; empty) and a fully uncensored window. Nine of the twelve possible cells were run; the three missing cells pair a start position with a window already covered by both of its neighbours. The table holds the model set fixed at the ten forecasters shared by every window; dropping Chronos, whose published forecasts condition on the 16-quarter context, moves the holding-dominated correlation from +0.591 to +0.455, so the design comparison is run on the common set. Chronos was also refit zero-shot on the 12-quarter context, which puts all eleven models in every arm; that comparison appears below and makes matters worse, not better.

Table 5: The fleet-level correlation under nine evaluation designs (ten forecasters common to all arms). The per-SKU τ barely moves; the method-mean correlation does not settle on a sign.

Start	Window	Order %	Stockout %	ρ pooled	ρ stockout	ρ holding	τ
S	1–16 → 17–20	18.2	9.1	−0.309	−0.285	+0.455	+0.405
mid	1–16 → 17–20	26.2	10.4	−0.309	−0.358	+0.455	+0.337
s	1–16 → 17–20	36.3	11.9	−0.309	−0.236	+0.455	+0.333
zero	1–16 → 17–20	72.3	36.2	−0.455	−0.273	+0.418	+0.435
S	1–12 → 13–16	18.9	34.0	+0.685	−0.212	−0.103	+0.411
mid	1–12 → 13–16	27.8	36.0	+0.600	−0.418	−0.103	+0.347
S	1–12 → 13–20	28.6	24.6	+0.345	+0.091	+1.000	+0.403
mid	1–12 → 13–20	36.2	25.7	+0.273	+0.055	+1.000	+0.345
zero	1–12 → 13–20	66.2	40.6	+0.127	−0.055	+0.927	+0.396

Two things are true at once. The opening position, which is the obvious worry (under a saturated start only 18.2 % of SKUs ever place an order, and cost is 90.5 % holding), barely matters: starting every SKU empty raises the ordering share to 72.3 %, lifts stockout cost to 36.2 % of the total, and moves the holding-dominated correlation from +0.455 to +0.418. The regime result is not an artifact of an unexercised policy. The window is another matter. The same statistic is −0.103 on train 1–12 test 13–16 and +1.000 when the test window extends to 13–20. The first comparison moves the test period along with the training length, so we attribute it to neither alone; the second holds training fixed and still spans most of the range a correlation can take. All three windows are uncensored, and any of them could have been the published design.

A SKU bootstrap within each arm settles what the swing means, and the answer is the less flattering of the two candidates. The per-arm intervals are wide: $[-0.21, +0.99]$ in the arm at −0.103 and $[+0.64, +1.00]$ in the arm at +1.000, and the extreme arms' intervals overlap. So the grid does not establish that design variation exceeds sampling variation. What it establishes is that the instrument is underpowered for the question: under one defensible design the statistic reads as confidently positive (bootstrap share of replicates at or below zero, 0.001), under another it is indistinguishable from zero (0.334), and the two readings cannot be told apart. The model set is not the cause; with the design held fixed, eleven models give +0.591 $[+0.45, +1.00]$ and ten give +0.455 $[+0.42, +1.00]$. And the refit Chronos widens the swing rather than closing it: with all eleven models in every arm, the same statistic runs from −0.236 (bootstrap share at or below zero, 0.447) to +1.000, and the extreme arms' intervals, $[-0.37, +0.99]$ and $[+0.61, +1.00]$, again overlap.

The $1,600 label is part of the problem too. It is a price split, held fixed while the realized cost mix moves across arms. Classifying each SKU instead by its realized cost mix within each arm puts 71 % to 93 % of SKUs in the holding-dominated group, because with these opening positions most SKUs never stock out, and it produces a different and equally unstable set of correlations, running from −0.588 to +0.818 across the same nine arms. Neither labelling yields a stable fleet-level statistic.

The quantity that does not care is the per-SKU τ, computed over 15,348 SKUs rather than ten means: it sits between +0.333 and +0.435 in every arm. The interval of $[+0.49, +1.00]$ quoted above is real, but it holds the design fixed, and the −0.103 arm sits far outside it while that arm's own interval spans $[-0.21, +0.99]$. Sampling and specification uncertainty are both large here, and the honest statement is that a rank correlation over ten or eleven method means is underpowered for this question under any design.

5.5 What to use instead

The obvious remedy does not work. If MAE targets the wrong fractile, score each SKU at its own q^* instead. We tested that and it fails. Pinball loss at q^* is confounded with level just as badly in the opposite direction: the per-SKU rank correlation between that loss and forecast level has median $-\mathbf{1.000}$, and fleet-level $\rho(\text{pinball}, \text{cost})$ is $-\mathbf{0.491}$ against MAE's +0.018. It helps only where holding dominates (+0.809, $p = 0.003$). Swapping the metric swaps the direction of the confound rather than removing it.

The reason is structural. Under the model-agnostic σ the policy sees the forecast only through the scalar $\bar{D}$, so simulated cost is a deterministic function of level and *no* point-error metric can add information about cost. There is no second channel to exploit.

A distributional policy does open one, but only just. Under the native-quantile policy the reorder point is $s = \text{LT} \cdot Q_\alpha$, so the policy reads a quantile rather than a mean. That is a genuine second channel: the per-SKU rank correlation between level and Q_α has median +0.738, with $|\rho| > 0.99$ on only 14.4 % of SKUs, and cost tracks Q_α (median $\rho = +1.000$) more tightly than level (+0.816).

Table 6: Which metric predicts native-quantile cost (per-SKU median rank correlation, demand-active SKUs).

Metric	median ρ with cost
MAE	+0.632
Pinball at the service level	**+0.800**
CRPS over the five fitted quantiles	+0.400

Those are medians of per-SKU rank correlations, and each metric's correlation is defined on a different SKU subset, so the referee-proof version restricts all three to the 4,824 demand-active SKUs where every correlation exists and pairs the differences within SKU. The medians are unchanged there. The paired differences resolve less than the medians suggest: pinball-at-service-level minus MAE has mean +0.024 with interval $[-0.003, +0.051]$, touching zero, so the advantage over MAE is directional rather than established. What is established is the negative half: CRPS minus MAE is -0.107 with interval $[-0.133, -0.082]$, excluding zero. A generic distributional score is strictly worse than the point metric it was meant to improve on, because CRPS averages over the whole quantile grid including the median region the decision never reads; the loss at the fractile the policy consumes is at worst MAE's equal and directionally better. This is Gneiting's (2011) consistency principle observed inside a deployed policy chain: a scoring function should be consistent for the functional the decision reads, and the failure of an unweighted CRPS here is the case for quantile-weighted scoring made by Gneiting & Ranjan (2011). Only four models emit quantiles, so the fleet-level version of this comparison rests on four points and we do not interpret it.

5.6 Does the confound generalise? A test on public M5 data

If the confound were a property of intermittent demand it would reappear on the M5 competition data. It does not. Across 30,490 M5 series and eight classical forecasters, the per-series rank correlation between forecast level and MAE has median +**0.194**, with 1.5 % of series at exactly +1, against median +1.000 and 78.6 % here. By demand class the M5 medians are ordered as expected (Smooth +0.157, Erratic +0.229, Intermittent +0.253, Lumpy +0.301) but none is close to the maritime result.

The analytic condition explains the gap. Median demand is zero over the out-of-sample window for **10.1 %** of M5 series against **77.3 %** here. So the difference is how zero-dense the evaluation window is, not the horizon length and not the Syntetos-Boylan class label. Conditioning on the analytic condition, the mechanism does reappear on M5: median rank correlation +**0.614** on the 3,082 series where median demand is zero, against +**0.157** on the 27,384 where it is not.

The claim is therefore conditional and its severity is measurable in advance from the evaluation window alone. That also reconciles a disagreement rather than adding to it: competition-data studies and spare-parts studies sit on opposite sides of this condition.

5.7 Predictive-quantile policies

Table 7: Fill-matched cost change of the native-quantile policy against the normal-approximation policy, with achieved 99th-percentile coverage.

Model	κ to match fill	Q99 coverage	cost Δ at matched fill
ZIP	0.88	0.963	−14.8 %
HNB	0.91	0.973	+1.5 %
LGBM	0.51	0.995	−64.4 %
Chronos	3.83	0.961	+4.2 %

All four reach the target fill. The two models whose upper tails are best calibrated at the nominal 99 % level, LightGBM (0.995) and HNB (0.973), sit at opposite ends of the cost outcome (Figure 4, Figure 5), so calibration at a single level does not by itself determine whether a predictive-quantile policy pays. The ZIP saving is stable across sampling seeds (mean −15.2 %, range −15.5 % to

−14.8 % over five seeds). A global multiplicative conformal correction moves every one of these by less than half a percentage point, so it neither rescues nor damages any model here.

We make no general claim from this table. The honest summary is that a predictive-quantile policy helped two of four models on this data and was approximately neutral for the other two, and that we cannot separate the effect of tail calibration from the effect of the fill-matching procedure itself.

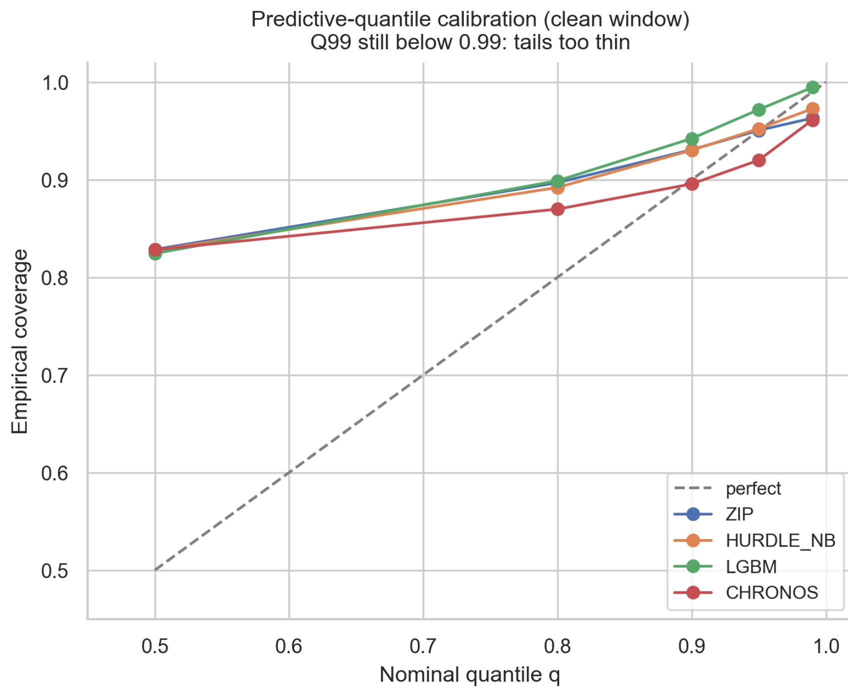

Figure 4: Predictive-quantile calibration on the evaluation window.

5.8 Deployed policy comparison

On the 1,579 SKUs where the deployed parameter set can be matched, ten of the eleven model-driven policies beat the deployed policy on cost. The best, HNB, costs \$567 against \$1,151, a 51 % reduction (Figure 6). LightGBM is the exception at \$1,543, or 34 % more expensive than deployed, because its point forecast over-stocks and this overlap is richer in demand-active SKUs than the sample as a whole.

The direction is robust and now has an interval: a paired bootstrap over the overlap SKUs puts the gap at −50.7 % with a 95 % interval of [$\mathbf{-60.2}$ %, $\mathbf{-41.0}$ %], comfortably excluding zero. The cheapest model in each cell beats deployed in 80 of 80 cost-configuration cells and in all four service-level scenarios, by 49 % to 53 %. Which model is cheapest is configuration-dependent (IMAPA wins 31 cells, Chronos 23, HNB 13, ZIP 9, ADIDA and SBA 2 each), so there is no single best forecaster to crown.

We do not report the magnitude with confidence. The data contain no realised stockouts, so the simulator cannot be validated against ground truth. Against the recorded inventory snapshot it tracks in shape but not in level: simulated deployed on-hand correlates with recorded stock at +0.663 in logs but sits at about half its level (median ratio 0.50), and the simulated deployed fill of 0.864 falls short of the operator's stated 0.95 to 0.99 goal. The simulator under-stocks relative

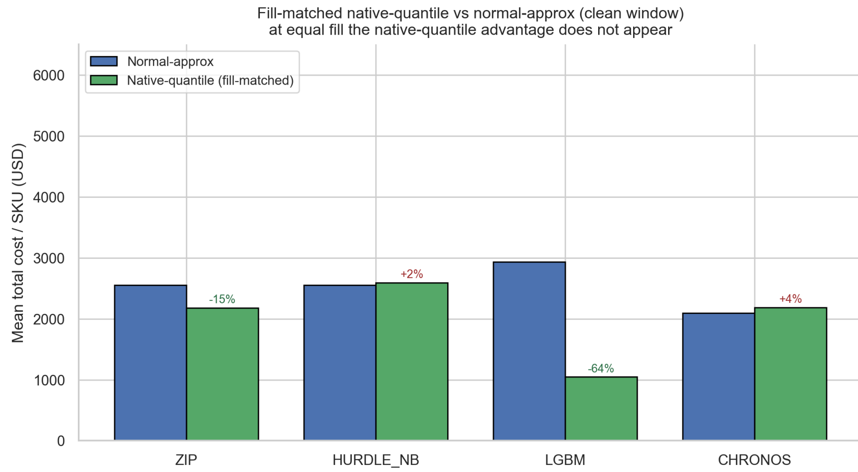

Figure 5: Fill-matched native-quantile versus normal-approximation cost.

to reality. Model-versus-model comparisons share the simulator and are unaffected; the absolute deployed figures are directional. The overlap is also a biased subsample, over-weighted toward lumpy and high-ABC parts. We do not compare fill on the overlap, because model fill was computed sample-wide rather than on this subset.

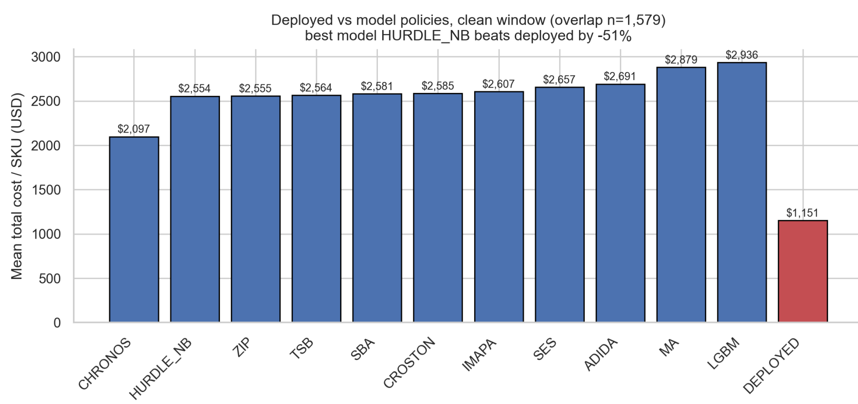

Figure 6: Deployed versus model-driven policy cost on the overlap SKUs.

5.9 A negative result: censoring does not reorder forecasters

Because the recent quarters are incomplete, an obvious worry is that a test window placed there would flatter models that under-forecast. We tested it. Holding the training window (quarters 1 to 16) and the horizon (four quarters) fixed, and moving only the test window into the incomplete region (quarters 25 to 28), the demand-active fraction falls from 38.6 % to 11.4 %, confirming the censoring is real and severe. The cost ranking does not move: the ordering of all ten models is identical across the two windows, the moving average is ninth of ten in both, and the within-SKU τ is unchanged (+0.405 clean against +0.426 censored, overlapping intervals).

A ranking change does appear if one compares the clean window against the original naive split (train quarters 1 to 20, test 21 to 28), where the moving average is the cheapest model. But that comparison moves the training window and doubles the horizon as well as shifting the test period, so it cannot be attributed to censoring. We report this because we initially believed the opposite, and because the controlled version of the experiment is cheap and we suspect other cost-aware studies have not run it.

5.10 Two design checks: information sets and design weights

Two things about the setup would not survive a referee. Three of the eleven forecasters saw less history than the rest, and the sample is not the fleet whose means the paper reports. We fixed both and report what moved.

Equal information sets. Croston, SBA and TSB update their smoothed state from the previous observation, so reading the forecast at the last training column returns a forecast whose state has consumed only fifteen of the sixteen training quarters. The final quarter reaches it only through the initialisation constants, whose weight decays as $(1-\alpha)^T$. Every other forecaster here conditions on the full window: SES takes an explicit extra step, the moving average reads the last two quarters directly, ADIDA and IMAPA end with a smoothing step on the most recent aggregated block, ZIP and Hurdle-NB fit moments over all sixteen columns, and LightGBM and Chronos take lags and context ending at the last quarter. Three of eleven models were being scored on a shorter history than their competitors.

Running one further iteration of the same recurrence removes the asymmetry. The three affected models improve, by 0.36 %, 0.37 % and 2.12 % of MAE and 0.30 % to 0.56 % of cost, and the other eight are bit-identical, which is the check that the change touched nothing else. No conclusion moves. Chronos remains both the most accurate and the cheapest; the holding-dominated correlation is unchanged at +0.591; the within-SKU τ is +0.387 either way; the deployed gap moves from −50.7 % to −50.9 %. Two labels do change: TSB rises from fourth to second cheapest, and it replaces Hurdle-NB as the cheapest model on the deployed overlap. Since Section 5.2 already establishes that adjacent models are not separable, neither label carries an inferential claim.

Design weights. Sampling fractions run from 1.0 % to 100 %, so the Horvitz–Thompson estimate carrying $w_h = N_h/n_h$ answers a different question from the raw mean: what the fleet looks like, rather than what the sample does. The two differ substantially.

Table 8: Sample estimates against Horvitz–Thompson population estimates under the sampling design weights. Every dollar figure in this paper is a sample figure; the fleet is markedly more intermittent than the sample.

Quantity	Sample	Population
Mean cost per SKU (Chronos)	\$2,097	\$434
Demand-active SKU share	38.6 %	14.8 %
Holding-dominated SKU share	3.2 %	6.0 %
Cost share in holding-dominated	82.3 %	62.1 %
Mean within-SKU τ (demand-active)	+0.387	+0.436
ρ(MAE, cost), holding-dominated	+0.591	**+0.991**
ρ(MAE, cost), stockout-dominated	−0.345	**−0.882**
ρ(MAE, cost), pooled	+0.018	+0.109
Deployed gap	−50.7 %	−52.1 %

Every dollar figure in this paper is therefore a sample figure, and the fleet is far more intermittent than the sample: weighted, only 14.8 % of SKUs see demand in the test window. Chronos remains the cheapest model under weighting and the deployed gap is unchanged in direction and size.

The regime contrast sharpens rather than dissolving, which is convenient enough that it deserves scrutiny. A weighted correlation can be one heavy stratum wearing a large weight, so we decomposed it. The holding-dominated result is robust: it is positive within every major stratum (+0.845, +0.864, +0.809, +0.864 for the four largest by weight); removing any of the four largest strata leaves it above +0.97, and the full leave-one-stratum-out minimum is +0.69, at a stratum carrying 0.5 % of the weight. The stockout-dominated result is not robust in this way. It falls from −0.882 to −0.182 when the Intermittent × C cell is removed, and the within-stratum values have no consistent sign, running from −0.709 to +0.936. That cell is 68 % of the fleet by weight, so −0.882 is a true statement about this population and a statement about one stratum at the same time.

So the contrast survives both checks and the magnitude of its negative half does not. The correlation is positive where holding cost dominates, in the sample and in the population alike; Section 5.3 says what it is made of. Where stockouts dominate it does not, and how negative the relationship looks depends on the stratum. Kish's effective sample size under these weights is 3,127 overall and 162 in the holding-dominated group, so the weighted intervals are wide by construction, and we report them alongside the unweighted ones rather than in place of them. These are the two checks the regime contrast survives; the window sensitivity of Section 5.4 is the one it does not.

6 Discussion

The results cohere around one point: the fleet-level correlation between method-mean accuracy and method-mean cost, which is how this literature usually asks whether accuracy matters, does not measure what it appears to measure on intermittent data, and the failure has structure.

It is confounded wherever the zero-median condition binds. MAE-rank is level-rank there, cost decomposes into a component level raises and a component level lowers, and the cost regime picks the sign. And it is fragile even setting the confound aside: eleven points under one design, moving across most of the admissible range under defensible design changes, with per-design intervals too wide to tell the movement from noise. Neither failure is visible from inside a single study, which may be part of why two decades of asking the question have not produced a consistent answer.

What should an accuracy-versus-cost study report instead? Four things, all cheap.

First, the zero-median share of its evaluation window. That number determines whether a median-targeting metric can separate forecast quality from forecast level at all: at 77.3 % it cannot, at M5's 10.1 % it largely can. It is computable before any model is fit.

Second, a scale-family placebo. Score k-scaled copies of one forecaster with the study's own pipeline. Whatever accuracy-cost relationship the pipeline awards them is its level effect, and only signal beyond that is evidence about accuracy. Ours awards them +1.000 where holding cost dominates.

Third, per-SKU paired statistics alongside the method-mean correlation, and the correlation's movement across at least one alternative window. A sampling interval on a fixed design is only part of the uncertainty.

Fourth, the critical fractile. $q^* = C_u/(C_u + C_o)$ locates the regime boundary analytically, at \$1,600 here with a median q^* of 0.981, and where a distributional forecast is available it should be scored at the fractile the policy reads, not with a generic proper score, which we find performs worse than MAE in a four-model comparison.

None of this says accuracy is worthless. The within-SKU τ of +0.387 among demand-active SKUs is stable across all nine evaluation designs and consistent with better forecasts costing less for a given part. What fails is the instrument that aggregates this into "the accurate model is the cheap model". And the deployed-policy comparison shows where the money is: the safety-stock parameterisation separates the deployed policy from the model-driven alternatives by far more than the forecasters separate from each other.

7 Limitations

1. **Four-quarter test horizon**, forced by the data vintage. Short for intermittent demand.
2. **One operator, one segment.** External validity is unestablished; this is a replication on one new domain, not a general law.
3. **The design weights are large and the effective sample is small.** Section 5.10 reports Horvitz–Thompson estimates alongside the unweighted ones, so fleet claims are no longer read off a raw sample mean. But w_h reaches 96, Kish's effective n is 3,127, and in the holding-dominated group it is 162. Weighted intervals are correspondingly wide, and the weighted stockout-dominated correlation is carried by a single stratum.
4. **Cost concentration, and what it costs us in power.** 3.2 % of SKUs carry 82.3 % of simulated cost, so sample-mean cost is effectively a statement about a small high-value subset. Under a paired bootstrap only one of ten models is distinguishable from the cheapest, so the cost ranking should be read as descriptive. A design that trims or stratifies the tail would have more power, at the cost of no longer describing actual spend.
5. **Simulator not validated against realised stockouts**, because the data contain none. It under-stocks by roughly 2× against the recorded snapshot.
6. **Lost-sales rather than backorder dynamics**, which suits consumable spares but not repairable or critical items where demand genuinely queues.
7. **The window is the exposure, not the opening position.** With on-hand initialised at S, 81.8 % of SKUs never reorder within the four test quarters. The specification grid of Section 5.4 tests this directly: opening positions from saturated to empty move the holding-dominated correlation by less than 0.04, while the choice of evaluation window moves it across most of its range. What remains unresolved is which window is the right one, and eleven method means cannot settle that.
8. **Cost is 90.5 % holding**, so the cost ranking is close to a bias ranking and does not isolate the value of a forecast's shape.
9. **Assumed cost parameters**, swept but not operator-supplied. The regime crossover at $1,600 is a consequence of the assumed $100 stockout penalty; a different penalty moves the boundary, though not the qualitative result.
10. **Model-agnostic σ_D**, which isolates point-forecast quality but does not test forecaster-specific uncertainty.

11. **Untuned baselines.** ZIP and HNB are intercept-only and share a point forecast; LightGBM is recursive multi-step; Chronos is zero-shot.
12. **The parameters are inherited, not fitted.** The information-set asymmetry that used to sit here is fixed in Section 5.9. What remains is that the smoothing constants for SBA, Croston, TSB, SES and the aggregate methods come from the operator's deployed configuration rather than being optimised on the training window, so these methods are evaluated as deployed rather than at their best.
13. **Coarse granularity**: quarterly review, integer-quarter lead times, and a cycle-service z applied to criticality tiers specified as availability targets.
14. **The distributional comparison rests on four models.** Only ZIP, Hurdle-NB, LightGBM and Chronos emit quantiles, so the fleet-level version of Section 5.5's metric comparison has four points and is not interpreted.
15. **The M5 comparison is a metric test, not a cost test.** We reuse per-series accuracy and bias for eight classical methods computed in a companion study on the public M5 data, and recover forecast level from bias; no inventory simulation is run on M5, so Section 5.6 establishes the confound's prevalence there but not its cost consequences.

8 Conclusion

On 15,348 SKUs of real maritime spare-parts data, the statistic this literature uses to ask whether forecast accuracy predicts inventory cost turns out, under policies that read the forecast through its mean, to measure forecast level. MAE is minimised at the median of realised demand, which is zero for 77.3 % of these SKUs, so MAE-rank is level-rank by identity, with per-SKU rank correlation median +1.000; six scaled copies of one forecaster, containing identical information about demand, reproduce the full regime pattern at +1.000 where holding cost dominates. The boundary between regimes is the newsvendor critical fractile crossing the median, at a derivable unit price of \$1,600, with the correlation's sign differing across that price as predicted. The instrument is fragile besides: across nine defensible evaluation designs the holding-dominated correlation runs from -0.103 to $+1.000$, while the per-SKU τ stays between $+0.333$ and $+0.435$. Scoring at q^* inherits the confound with the sign reversed; the fractile the replenishment decision reads escapes it, and a generic proper score is strictly worse than MAE. On public M5 data the zero-median condition holds for 10.1 % of series against 77.3 % here, which helps reconcile why competition-data and spare-parts studies disagree about whether accuracy matters. Three secondary results: ten of eleven model-driven policies beat the deployed parameterisation, direction bounded away from zero ($[-60.2\,\%, -41.0\,\%]$); a zero-shot foundation model is the most accurate and cheapest point estimate but separable from almost none of its competitors; and censoring of the most recent quarters, though real and severe, does not reorder forecasters. The design checks of Section 5.10, equalised information sets and design weights, leave these conclusions standing. What they do not leave standing is any temptation to read an eleven-point correlation as a stable fact about the world.

Reproducibility and analysis note

The full pipeline (anonymisation, sampling, all eleven forecasters, the (s, S) simulator and every robustness check) is released as code, and the anonymised data are released with it. Results were produced under a claims-verification script that cross-checks every number in this paper against the analysis ledgers; it is part of the release.

This paper went through an adversarial review that changed its conclusions, and the corrections are worth recording because each one is a plausible failure mode for a cost-aware forecasting study.

1. **A predictive-quantile composition bug.** LightGBM's quantiles were composed as the conditional-size quantile multiplied by an indicator that the occurrence probability exceeded 0.5. That collapsed the entire predictive distribution to a point mass at zero for two thirds of the sample and left the rest on an un-remapped conditional quantile. Corrected to the hurdle mixture, LightGBM's Q99 coverage moved from 0.941 to 0.995 and its fill-matched cost outcome from +204 % to −64.4 %. An earlier draft reported the buggy figure as evidence that machine-learning models have tails too thin for inventory use.

2. **Monte Carlo sample size masquerading as miscalibration.** Chronos was sampled 30 times while ZIP and HNB were sampled 1,000 times, which biases the empirical 0.99 quantile downward. At 1,000 seeded draws its Q99 coverage moved from 0.939 to 0.961 and it reached target fill at $\kappa = 3.83$ rather than failing at the search bound.

3. **Non-deterministic training, in two layers.** LightGBM produced MAE of 8.526, 8.555 and 8.987 on three runs at the same seed. The first layer was the model: its stochastic parameters were unseeded and LightGBM is non-deterministic when multithreaded, so we pinned the seeds, `deterministic`, `force_row_wise` and the thread count. That was not sufficient. The second layer was the data: Polars `unique()` does not guarantee row order, and LightGBM's bagging samples rows by position, so a reshuffled frame still produced a different model. Only after sorting the analysis frame to a canonical order did two separate processes agree bit-for-bit. Every number here is from the sorted pipeline; the regime correlation in particular moved from +0.673 ($p = 0.023$) to +0.591 ($p = 0.056$), which is why we do not lean on it.

4. **An unsupported headline.** An earlier draft claimed censoring flipped the cheapest forecaster to the most expensive. No artifact supported it, and the controlled experiment in Section 5.9 shows it is false.

5. **An over-strong dominance claim.** An earlier draft stated the deployed policy was beaten by every model-driven policy. LightGBM does not beat it.

6. **A vacuous robustness check.** An earlier draft offered MASE as corroboration of the MAE-based rank statistic. MASE is a positive per-SKU rescaling of MAE, so the statistic is identical by construction.

7. **A statistic with too little power.** The earliest version measured the accuracy-cost relationship as a Spearman correlation over eleven forecasters, whose bootstrap interval spanned most of $[-1, 1]$.

8. **An unfair comparison.** SBA, Croston and TSB were scored on fifteen training quarters against sixteen for the other eight models, because their state update lags the observation by one period. Section 5.10 equalises it. All three improve and nothing else moves.

9. **A bootstrap interval quoted more precisely than it was estimated.** At $B = 2{,}000$ the holding-dominated lower bound ranged from +0.409 to +0.545 across seeds, and an earlier draft quoted +0.527 from one of them. B is now 20,000, the interval is quoted to two decimals, and the stable statistic, the bootstrap share of replicates at or below zero (0.0014), is reported with it.

10. **Fleet claims read off an equal-allocation sample.** Design weights reach 96 : 1 here, and the unweighted mean cost is roughly six times the population estimate. Section 5.10 reports both.

11. **Sampling-only uncertainty presented as the uncertainty.** An earlier draft gave the holding-dominated correlation a bootstrap interval of $[+0.49, +1.00]$ and treated that as the finding's uncertainty. The interval resamples SKUs with the evaluation design held fixed. Varying the design instead (Section 5.4) moves the same statistic from -0.103 to $+1.000$, and the per-arm intervals are wider still, so the quoted interval understated the total uncertainty from both sources.

12. **A flagship comparison quoted on non-identical SKU subsets.** The metric ordering of Section 5.5 compared medians over different SKU sets with no uncertainty attached. On the common set, paired inference establishes only the CRPS half; the pinball advantage over MAE is directional.

One earlier discrepancy is now resolved. A previous draft reported the Hurdle-NB fill-matched delta as $+2.1$ % in one script and -0.5 % in another, on the same window and configuration, and flagged the inconsistency rather than choosing. The non-determinism fixes of item 3 were the cause: after the canonical-ordering re-run the two scripts agree at $+1.5$ %, the value in Section 5.7's table.

References

Ansari, A. F., et al. (2024). *Chronos: Learning the Language of Time Series.* Transactions on Machine Learning Research. arXiv:2403.07815.

Bandara, K., Bergmeir, C., & Smyl, S. (2020). Forecasting across time series databases using recurrent neural networks on groups of similar series: a clustering approach. *Expert Systems with Applications*, 140, 112896.

Croston, J. D. (1972). Forecasting and stock control for intermittent demands. *Operational Research Quarterly*, 23(3), 289–303.

Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, 106(494), 746–762.

Gneiting, T., Balabdaoui, F., & Raftery, A. E. (2007). Probabilistic forecasts, calibration and sharpness. *JRSS B*, 69(2), 243–268.

Gneiting, T., & Ranjan, R. (2011). Comparing density forecasts using threshold- and quantile-weighted scoring rules. *Journal of Business & Economic Statistics*, 29(3), 411–422.

Hyndman, R. J., & Koehler, A. B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4), 679–688.

Ke, G., et al. (2017). LightGBM: A highly efficient gradient boosting decision tree. *NeurIPS* 30.

Kolassa, S. (2016). Evaluating predictive count data distributions in retail sales forecasting. *International Journal of Forecasting*, 32(3), 788–803.

Kolassa, S. (2020). Why the "best" point forecast depends on the error or accuracy measure. *International Journal of Forecasting*, 36(1), 208–211.

Kourentzes, N. (2014). On intermittent demand model optimisation and selection. *International Journal of Production Economics*, 156, 180–190.

Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2022). The M5 accuracy competition: results, findings, and conclusions. *International Journal of Forecasting*, 38(4), 1346–1364.

Nikolopoulos, K., Syntetos, A. A., Boylan, J. E., Petropoulos, F., & Assimakopoulos, V. (2011). An aggregate–disaggregate intermittent demand approach (ADIDA) to forecasting: an empirical proposition and analysis. *JORS*, 62(3), 544–554.

Petropoulos, F., Wang, X., & Disney, S. M. (2019). The inventory performance of forecasting methods: evidence from the M3 competition data. *International Journal of Forecasting*, 35(1), 251–265.

Prak, D., & Teunter, R. (2019). A general method for addressing forecasting uncertainty in inventory models. *International Journal of Forecasting*, 35(1), 224–238.

Salinas, D., Flunkert, V., Gasthaus, J., & Januschowski, T. (2020). DeepAR: probabilistic forecasting with autoregressive recurrent networks. *International Journal of Forecasting*, 36(3), 1181–1191.

Silver, E. A., Pyke, D. F., & Thomas, D. J. (2017). *Inventory and Production Management in Supply Chains* (4th ed.). CRC Press.

Snyder, L. V., & Shen, Z.-J. M. (2019). *Fundamentals of Supply Chain Theory* (2nd ed.). Wiley.

Syntetos, A. A., & Boylan, J. E. (2005). The accuracy of intermittent demand estimates. *International Journal of Forecasting*, 21(2), 303–314.

Syntetos, A. A., & Boylan, J. E. (2006). On the stock control performance of intermittent demand estimators. *International Journal of Production Economics*, 103(1), 36–47.

Syntetos, A. A., Nikolopoulos, K., & Boylan, J. E. (2010). Judging the judges through accuracy-implication metrics: The case of inventory forecasting. *International Journal of Forecasting*, 26(1), 134–143.

Teunter, R. H., Babai, M. Z., & Syntetos, A. A. (2010). ABC classification: service levels and inventory costs. *Production and Operations Management*, 19(3), 343–352.

Teunter, R. H., & Duncan, L. (2009). Forecasting intermittent demand: a comparative study. *Journal of the Operational Research Society*, 60(3), 321–329.

Teunter, R. H., Syntetos, A. A., & Babai, M. Z. (2011). Intermittent demand: linking forecasting to inventory obsolescence. *European Journal of Operational Research*, 214(3), 606–615.

Theodorou, E., Spiliotis, E., & Assimakopoulos, V. (2025). Forecast accuracy and inventory performance: insights on their relationship from the M5 competition data. *European Journal of Operational Research*, 322(2), 414–426. doi:10.1016/j.ejor.2024.12.033.

Vandeput, N. (2020). *Inventory Optimization: Models and Simulations.* De Gruyter.

Wallström, P., & Segerstedt, A. (2010). Evaluation of forecasting error measurements and techniques for intermittent demand. *International Journal of Production Economics*, 128(2), 625–636.